

EXERCISE SHEET 2

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

2.1. Let G be a group containing an element of finite order $n > 1$ and exactly two conjugacy classes. Prove that $|G| = 2$. [Hint: Show first that n must be prime.]

2.2. Prove that every group of order p^2 , for p a prime, is abelian.

2.3. Let G be a finite p -group, for a prime p . If $N \trianglelefteq G$ is a normal subgroup of order p , show that N is contained in $Z(G)$.

EXTRA QUESTIONS

2.4. One can check that

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mid b \in \mathbb{Z}_7, a \in \{1, 2, 4\} \right\}$$

is a group of order 21. Regard

$$X = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \mid x, y \in \mathbb{Z}_7 \right\}$$

and $X \times X$ as G -sets in a natural way (via ordinary matrix multiplication). Find the orbit of $\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right) \in X \times X$.

2.5. Show that the alternating groups A_2 and A_3 are simple, but that A_4 is not simple.

2.6. As was done for S_3 in Lecture 1, confirm the class equation for S_4 .